Per-site ΔRMSE from adding mortality — C+T vs C+T+M

Below 0 = mortality reduces the site's RMSE | bars coloured by relative deadwood class (site maximum) | repeated CV (3×80 %) | tree cover ≥30 % (101 sites, 361 site-years) · deadtrees v5

ΔRMSE = RMSE(with M) − RMSE(without M)

GBoost (Optuna) · 12r | **GBoost (Optuna) · 24r** | **ghtGBM (Optuna) · 12** | **ghtGBM (Optuna) · 24** | **toGluon (ensemble) · 1** | **toGluon (ensemble) · 2**

GPPsat (µmol m⁻² s⁻¹)

57% improved | 58% improved | 56% improved | 55% improved | 57% improved | 64% improved

2.5 / 0.0 / -2.5 / -5.0

NEPmax (µmol m⁻² s⁻¹)

68% improved | 70% improved | 74% improved | 68% improved | 69% improved | 65% improved

2 / 0 / -2 / -4

ETmax (mm d⁻¹)

58% improved | 64% improved | 58% improved | 57% improved | 57% improved | 62% improved

0.02 / 0.00 / -0.02 / -0.04

WUE (g C mm⁻¹)

53% improved | 58% improved | 53% improved | 51% improved | 55% improved | 48% improved

0.8 / 0.4 / 0.0 / -0.4

Sites (sorted by ΔRMSE within each panel)

Relative deadwood
site maximum, mean of 100-500 m buffers
(breaks: mean ± 0.5 SD)

- Low (< 7.5 %, 37 sites)
- Mid (7.5–17.5 %, 42 sites)
- High (> 17.5 %, 22 sites)